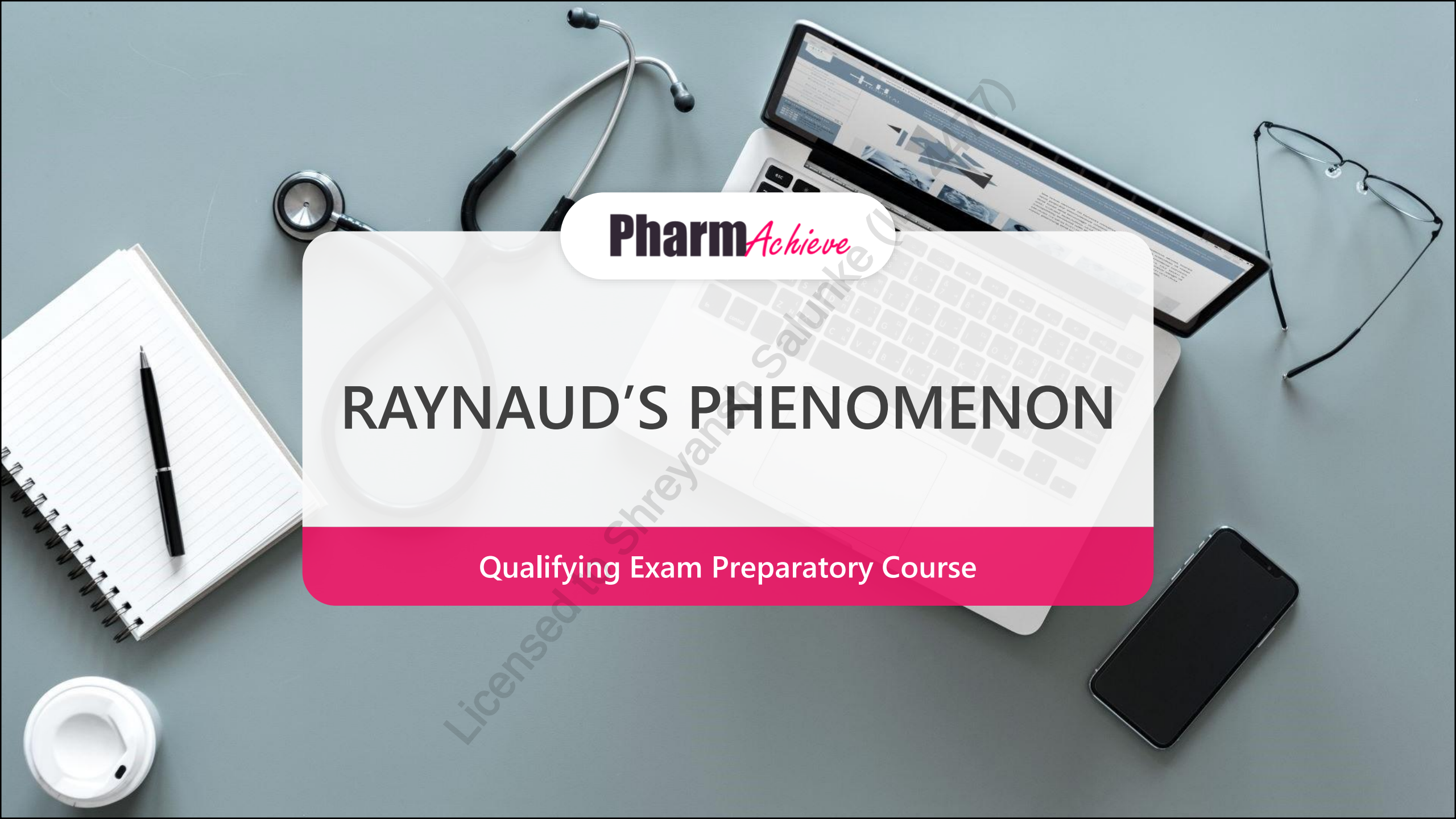


A top-down view of a grey desk with various items: a stethoscope, a laptop displaying a medical website, a tablet, a pair of glasses, a smartphone, a spiral notebook with a pen, and a white coffee cup. A large white pill-shaped graphic is centered over the laptop.

Pharm*Achieve*

RAYNAUD'S PHENOMENON

Qualifying Exam Preparatory Course

COMPETENCIES COVERED IN THIS PRESENTATION...

**Providing Care:
Clinical Care**

**Providing Care:
Distribution**

**Knowledge
& Expertise**

**Communication
& Collaboration**

**Leadership &
Stewardship**

Professionalism

LEGEND

ANA Antinuclear Antibody

BID Twice Daily

CCB Calcium Channel Blockers

CNS Central Nervous System

DHP Dihydropyridine

ENA Extractable Nuclear Antibodies

ER Extended-release

RF Rheumatoid Factor

RP Raynaud's Phenomenon

WHAT IS RAYNAUD'S PHENOMENON?

- Raynaud's is characterized by an extreme vascular response that discolours the affected area
- Generally, the discolouration follows a pattern of white to blue to red
- Raynaud's mainly affects the fingers and toes
 - Rarely it can affect the lips, nose, ears, and nipples
 - Typically affects one digit and spreads to the other digits

2 types of Raynaud's Phenomenon

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graph TD; A[2 types of Raynaud's Phenomenon] --> B[Primary Raynaud's phenomenon]; A --> C[Secondary Raynaud's phenomenon]; B --> D[Less severe]; B --> E[More common]; B --> F[Idiopathic]; C --> G[May be caused by an underlying disease]; C --> H[May be drug-induced];
```

Primary Raynaud's phenomenon

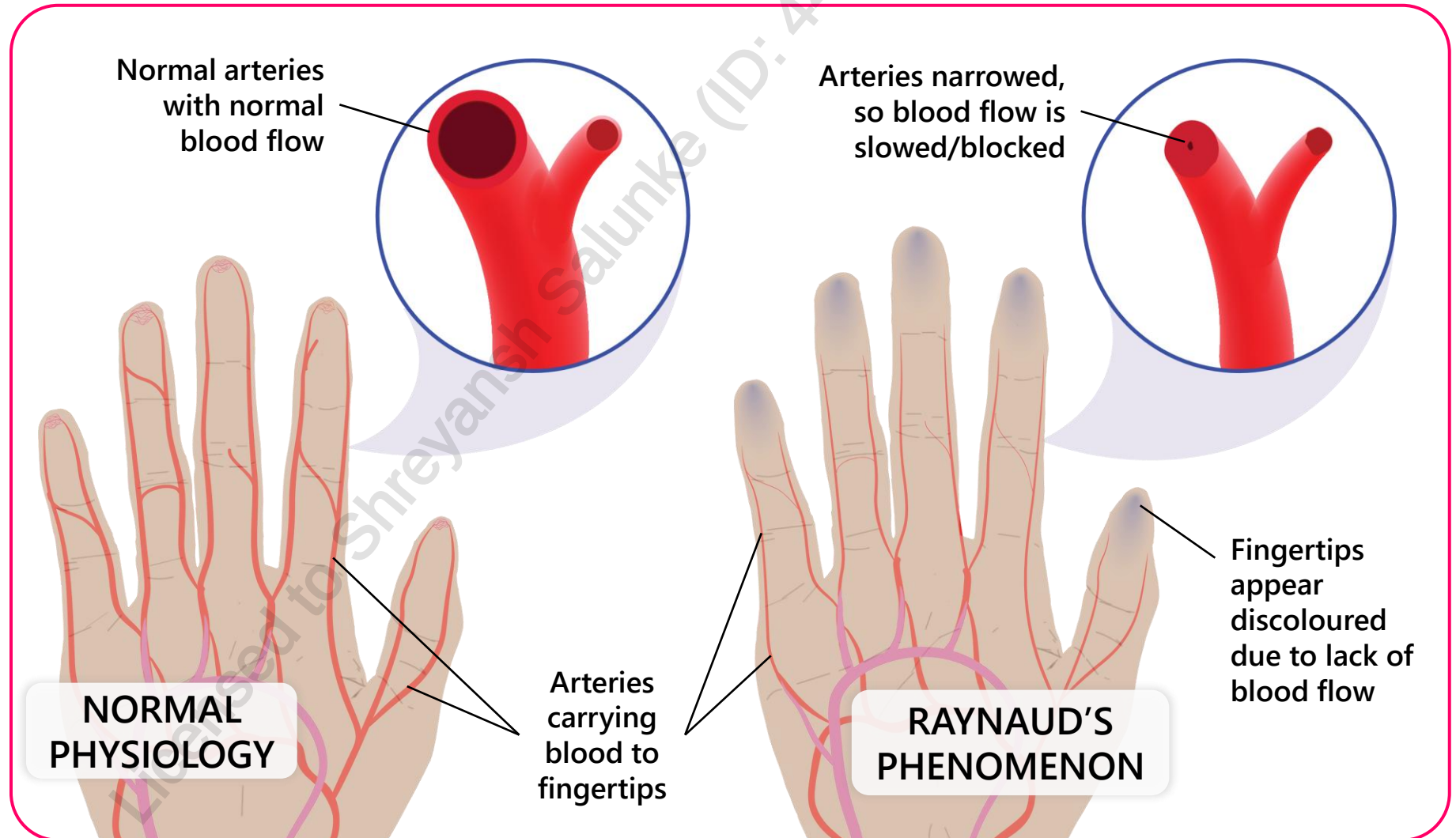
- Less severe
- More common
- Idiopathic

Secondary Raynaud's phenomenon

- May be caused by an underlying disease
- May be drug-induced

PATHOPHYSIOLOGY OF A RAYNAUD'S ATTACK

- Exaggerated vascular response causing arteries to narrow for a period
- Results in limited blood flow to digits (white / blue) lasting seconds-hours
- Followed by reactive hyperemia phase where blood flow returns to digits (red)



CLINICAL PRESENTATION

- The affected area may change colour and patient may feel throbbing, tingling, and burning
- Can start in one digit and then spread symmetrically in both hands
- More severe cases may result in skin sores, digital ulceration, gangrene
- Clinical presentation and medical history is sufficient for diagnosis
- Symptoms usually decrease in frequency and intensity during pregnancy

RISK FACTORS

General risk factors

- Females
- Family History
- Age < 30 years old
- Living in cold climate (e.g. frostbite)
- Repetitive damage to arteries and nerves (e.g, carpal tunnel)

Occupational/recreational hazards

- Exposure to some chemicals (vinyl chloride, nicotine)
- Vibration tools (jackhammer)
- Repetitive actions

Pharmacological agents

- Ergot derivatives
- Antineoplastics (e.g. bleomycin, cisplatin)
- Interferon alfa, beta
- CNS stimulants
- Cyclosporine
- Beta-blockers (unlikely, limited evidence)

Medical conditions

- Scleroderma
- Lupus
- Rheumatoid arthritis
- Atherosclerosis
- Sjogren's syndrome
- Hypothyroidism
- Pulmonary hypertension

GOALS OF THERAPY

- ✓ Minimize severity of signs and symptoms
- ✓ Confirm cause of secondary Raynaud's phenomenon
- ✓ Reduce and prevent the frequency of attacks
- ✓ Minimize adverse effects from drug therapy

NON-PHARMACOLOGICAL MEASURES

Avoid Triggers

- Discontinue offending medications if possible
- Lifestyle modifications: smoking cessation, reduce caffeine and alcohol intake
- Avoid vibrating tools and repetitive motions at work
- Stress reduction, good coping mechanisms and relaxation techniques

Managing temperature

- Reduce exposure to temperature extremes, especially the cold
- Warm affected areas slowly
- Wear protective clothing (e.g. gloves, warm socks, hat)
- Place hands/affected area under warm water

Physical activity

- Rotating arms like a windmill



Do not use
acupuncture due to
limited evidence of
efficacy

PHARMACOLOGICAL MEASURES

Drug class	Side effects	Comments
Dihydropyridine calcium channel blockers (e.g. nifedipine, amlodipine)	• Orthostatic hypotension • Headache • Dizziness • Flushing • Tachycardia (rare, DHPs) • Edema	• 1st line therapy • Take 30-60 minutes prior to being exposed to the cold • Take regularly during winter • Start at low doses and adjust every 4 weeks

PHARMACOLOGICAL MEASURES - ALTERNATIVES

FYI

- Alpha-blockers (e.g. prazosin)
- Nitroglycerin skin ointment
- Phosphodiesterase type 5 (PDE5) Inhibitors (e.g. sildenafil)
- Bosentan
- Sympathectomy (surgical method)
- Angiotensin Receptor Blocker (ARB) (e.g. candesartan)
- Selective Serotonin Reuptake Inhibitor (SSRI) (e.g. fluoxetine)

MONITORING PARAMETERS

EFFICACY ENDPOINTS		
Parameter	Degree of Change	Timeframe
Discolouration (white → blue → red)	Prevent or reduce frequency of episodes	Onset of action: few days-1 week
Throbbing, tingling, burning sensation		Dose adjustment: every 4 weeks
Skin sores, digital ulceration, gangrene	Prevent	Duration of therapy

SAFETY ENDPOINTS		
Parameter	Degree of Change	Timeframe
Adverse effects (orthostatic hypotension, headache, dizziness, flushing)	Prevent or minimize	Duration of therapy Monitor BP daily until dose is stable, then weekly thereafter

TAKEAWAY

Raynaud's Phenomenon is characterized by an extreme vascular response that presents with discolouration, pain, burning or tingling in the affected area

Clinical presentation and medical history is sufficient for diagnosis

Primary and secondary RP both respond to medications equally

Primary RP may be managed with non-pharmacologic measures alone and is the mainstay of therapy for pregnancy and breastfeeding

1st line pharmacological therapy is dihydropyridine calcium channel blockers

Daily medication administration will improve tolerance to side effects

CASE 1

NS, a 28-year-old female, presents to the pharmacy to ask for your advice. She states that she was outside playing in the snow with her children when she noticed her right index finger turning white. She said that it later turned blue, but after a while, it began burning and tingling and ultimately changed to red. Her past medical history is significant for attention deficit hyperactivity disorder (ADHD) and hypothyroidism. Her current medications include lisdexamfetamine 30 mg every morning and levothyroxine 125 mcg daily. She does not smoke or drink and is currently avoiding caffeine while breastfeeding her seven-month-old son.

CASE 1

1. How many risk factors for Raynaud's does NS have?
 - a) 3
 - b) 4
 - c) 5
 - d) 6

2. Which of the following is NOT an appropriate counseling point for NS?
 - a) Warm hands under hot water when coming inside
 - b) Wear protective clothing when going outside
 - c) Trying rotating arms like a windmill
 - d) Warm affected areas slowly

3. Which of the following is the MOST appropriate treatment option for NS?
 - a) Amlodipine
 - b) Candesartan
 - c) Diltiazem
 - d) Non-pharmacologic measures

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CHANGE LOG

May 2025

- Slide 4: Minor changes to the classification of Raynaud's phenomenon
- Grammar and formatting changes throughout
- Slide 13-14: Added case

July 2025

- Content reviewed; no changes made

September 2025

- Slide 2: Removed OD from legend per 'Do Not Use' list
- Slide 12: Added monitoring parameters

February 2026

- Slide 2: Changed NAPRA competencies
- Slide 9: Elaborated on stress reduction and protecting clothing
- Slide 10: Updated comments and added recommendation for dose adjustment every 4 weeks
- Slide 12: Elaborated on monitoring parameters